

TeleHopper: Simulating a Jumping Sensation as Proprioceptive Feedback for Teleportation in Virtual Reality via Electrical Muscle Stimulation

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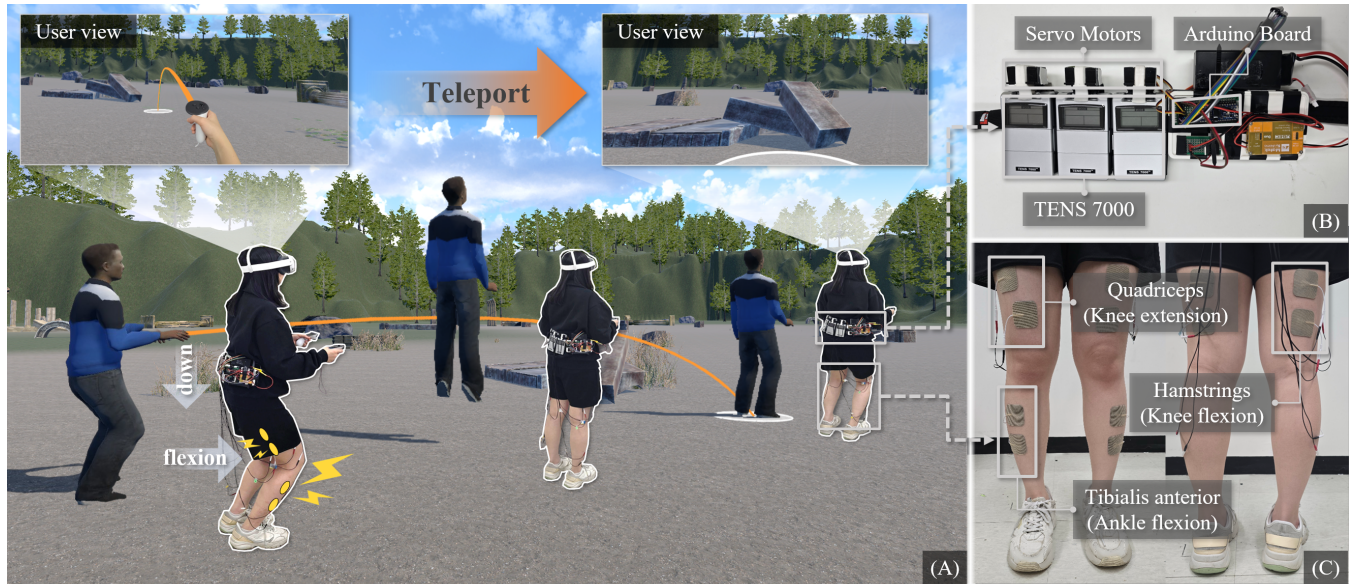


Figure 1: (A) Visualization of the proposed 'TeleHopper,' which uses Electrical Muscle Stimulation (EMS) on the lower body to provide proprioceptive feedback by inducing a jumping sensation during teleportation. (B) Depiction of the motorized EMS system used to apply feedback. (C) Depiction of electrodes attached to the leg muscles correlating with jumping motion.

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CHI EA '25, Yokohama, Japan
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ACM ISBN 979-8-4007-1395-8/25/04
<https://doi.org/10.1145/3706599.3720248>

Abstract

Teleportation, a method of instantly moving users to a target position, has become a widely adopted locomotion method in virtual reality. However, the lack of proprioceptive feedback for teleportation can diminish presence and increase workload, thereby limiting the overall user experience. In this study, we propose TeleHopper, a system that enhances the teleportation experience by simulating the sense of jumping during teleportation through Electrical

Muscle Stimulation-based haptic feedback. TeleHopper induces leg movements resembling a jumping motion and adjusts stimulation intensity based on travel distance, creating a realistic proprioceptive perception of leaping through space during teleportation. Experimental results evaluating TeleHopper's user experience showed a significant enhancement in sense of presence, as well as a significant reduction in mental workload. Through this study, we demonstrate TeleHopper's ability to deliver compelling proprioceptive feedback in teleportation, with varying stimulation intensity enhancing realism and aiding travel distance estimation.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; **Haptic devices**.

Keywords

Virtual Reality, Locomotion, Teleportation, Haptic, Electrical Muscle Stimulation

ACM Reference Format:

Juwon Um, Bocheon Gim, Seongjun Kang, Yumin Kang, Eunki Jeon, and SeungJun Kim. 2025. TeleHopper: Simulating a Jumping Sensation as Proprioceptive Feedback for Teleportation in Virtual Reality via Electrical Muscle Stimulation. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3706599.3720248>

1 Introduction

Virtual reality (VR) often accommodates virtual environments larger than the available physical tracking space [22, 41, 59], leading to the development of various locomotion methods to expand the range of virtual motion beyond real-world constraints [33, 63]. Teleportation is one such virtual locomotion method, designed for users to point a controller at a target destination and instantly move to the specified location [57]. It has become a widely accepted locomotion method in commercial VR applications [42, 57], offering advantages such as ease of use [6], rapid travel times [8, 31], and reduced motion sickness due to its discontinuous nature of movement [10, 61]. However, locomotion in real life is a multi-sensory activity that involves not only visual, vestibular, and tactile sensations, but also proprioception [49], a crucial component in perception of bodily motion [48]. In contrast, teleportation relies solely on visual cues, providing a limited sensory representation of locomotion [6, 57]. This resulting sensory misalignment can reduce presence [7, 10, 15, 64] and increase workload [7, 36], consequently diminishing the user's overall locomotion experience.

The absence of sensory cues in teleportation can be addressed by providing additional information through tactile feedback [9, 18, 57]. However, because teleportation is an experience that does not align with real-world locomotion, there is a need to actualize an imaginary sensation that could contextualize its instantaneous movement [7]. For example, Tseng et al. proposed HeadWind [57], which uses compressed air to represent the instantaneous motion of teleportation as a tactile wind sensation. However, challenges persist in addressing the absence of proprioception—an essential factor in movement perception [48] that enhances user embodiment and improves presence [25, 34].

To convincingly supplement a proprioceptive sensation to teleportation, our study focuses on the nature of teleportation as a visually perceived spatial "jump" [6, 57]. Past studies have described teleportation as a "jump" [1, 11, 41, 50] due to its similarities in abrupt transition to a designated location, with some directly correlating physical jumping as a method of teleportation input [20, 30]. Thus, we propose TeleHopper, a system that enhances the teleportation experience by simulating a jumping sensation through proprioceptive feedback, thereby evoking the feeling of leaping through space. To achieve this, we employ Electrical Muscle Stimulation (EMS), a lightweight and wearable technology that uses electrical impulses to elicit muscle contraction and subsequent kinesthetic actuation [37]. TeleHopper prompts users to prepare for a jump by bending their legs while pointing at the teleport destination, then induces leg extension to simulate leaping and ankle flexion to simulate landing. The intensity of TeleHopper's stimulation increases linearly with the teleport distance, mirroring the greater physical effort required for longer jumps.

In this paper, we explore the design of TeleHopper and evaluate its impact on the teleportation experience. In particular, this study aimed to investigate the impact of supplementing proprioception in teleportation on users' sense of presence and perceived workload associated with virtual locomotion, examined in a user study designed to evaluate TeleHopper through a virtual coin collection task. Additionally, post-interview feedback was collected to gather qualitative insights into the overall experience with TeleHopper, and to discuss potential directions for its future development in regards to its alignment with teleportation. The main contribution of this study is as follows: We propose a novel system that supplements the proprioception in teleportation by simulating the sensation of jumping through EMS, thereby enhancing the teleportation experience of VR users.

2 Related work

2.1 Teleportation in VR

VR employs diverse locomotion methods to facilitate navigation within expansive virtual environments beyond physical tracking spaces [16, 58, 59, 63]. In particular, Teleportation has been increasingly adopted in commercial VR applications [42, 57] due to its advantages compared to other locomotion methods, such as reduced motion sickness [10, 61] and effective manipulation [8]. However, the inherently discontinuous nature of teleportation also presents reported disadvantages, such as decreased sense of presence or spatial disorientation [2, 7]. Some studies have aimed to overcome these drawbacks, such as implementing a rapid, continuous transition to the target position [4].

Furthermore, there have been studies aimed at enhancing the teleportation experience by providing additional sensory information through tactile feedback. One such example is HeadWind [57], which simulates air resistance using compressed air and effectively enhanced realism, immersion, and enjoyment in VR gaming environments. Similarly, Han et al. proposed a method using a mid-air haptic system called Haptic Around [18], which simulates environmental changes before and after passing through a teleportation portal. Additionally, Chu et al. introduced MotionRing [9], a headband equipped with 360°-arranged vibrotactile feedback, and

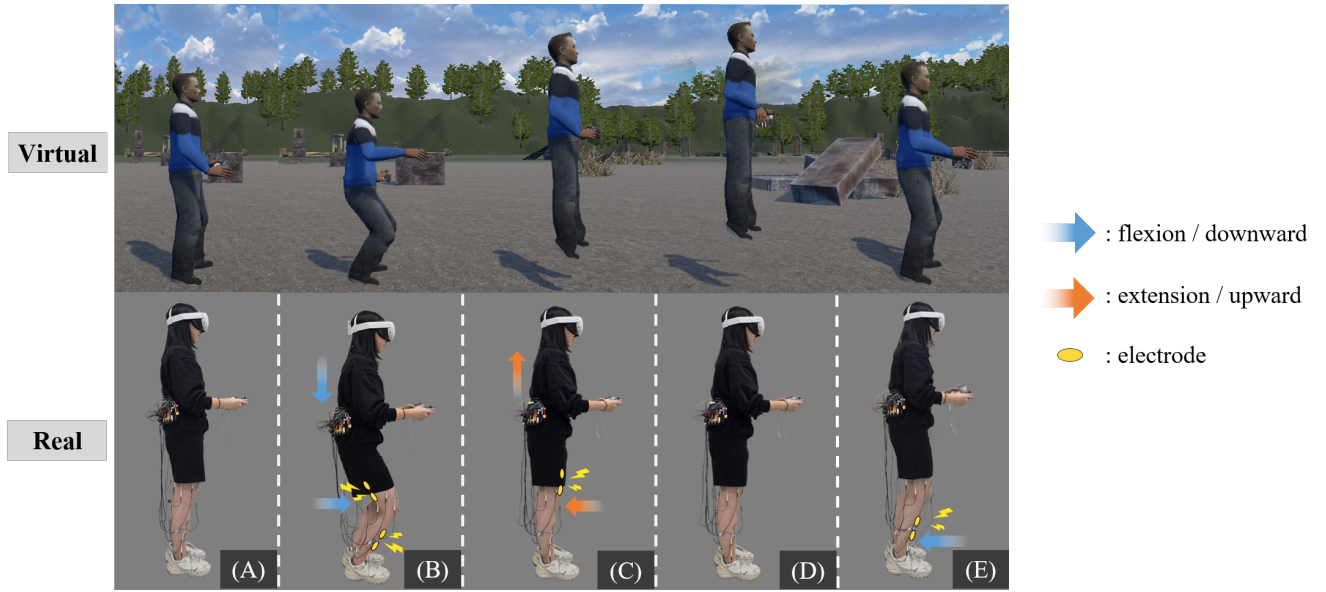


Figure 2: Sequential representation of TeleHopper’s EMS-based feedback design, simulating a jumping sensation during teleportation across the phases of (A) Idle-state, (B) Preparation, (C) Push-off, (D) Float, and (E) Landing.

suggested its application for simulating air resistance caused by sudden acceleration correlating with teleportation. As such, various tactile feedback methods have been proposed to contextualize bodily sensations to teleportation. However, studies that supplement proprioception, an essential element of locomotion sense, remain limited [48]. Therefore, this study aims to supplement the proprioceptive sensation in teleportation by simulating the sense of jumping through proprioceptive feedback.

2.2 EMS in VR

Haptic feedback in VR has the potential to strengthen immersion by conveying physical cues associated with virtual interactions [24, 39, 57]. EMS is one such technology, which delivers electrical signals through electrodes attached near muscles to induce muscle contractions [21, 55]. These contractions actuate the limbs involuntarily, so EMS has been used as a means to actuate proprioceptive feedback [37]. For instance, previous research simulated sensations of weight and resistance by contracting the opposing muscles to those used in interactions such as lifting or pulling virtual objects [27, 39]. Similarly, some studies simulated environmental sensations, such as the resistance felt while walking through water [23].

Other studies have utilized EMS to simulate physical responses resulting from interactions by contracting the muscles involved in actual motions. A representative example is the repulsion design, which replicates movements such as an arm being pushed back by the impact of punches in a boxing simulation [38] or shockwaves from an explosion [26]. These feedback mechanisms leverage EMS to induce user movements that correspond with virtual events, thereby providing users with realistic proprioception during interactions. In this context, this study employs EMS to simulate the sensation of jumping, aiming to supplement proprioception within the teleportation experience.

3 Implementation

The implementation of teleportation largely follows that of prior implementations [42, 57], where users designate the target destination using a controller, and discontinuous motion is applied during each teleportation. The maximum teleportation range was set to 10 meters [42, 51].

3.1 Hardware implementation and EMS personalization

To generate electrical stimulation for TeleHopper, we developed a wireless motorized EMS generator based on previous designs [12, 23] (Figure 1 (B)). The generator produces EMS signals at 100 Hz with a maximum current of 100 mA using the *TENS 7000* device. For remote control of stimulation intensity, motors were attached to each EMS channel, adjustable on a 0–99 scale via UDP communication over Wi-Fi connection. The device was mounted on the user’s lower back with a belt wrapped around the waist to guarantee user comfort and mobility. Furthermore, to determine the appropriate range of stimulation intensity that ensures safety while delivering a clear sensation of stimulation, we included an EMS personalization process. Specifically, a staircase protocol based on previous studies [12, 47] was employed to identify the motor threshold—the minimum intensity required to induce muscle contraction—and the pain threshold—the maximum intensity that does not cause pain. These values were set as the minimum and maximum intensity limits, respectively.

3.2 Feedback Design for TeleHopper

TeleHopper provides EMS-based haptic feedback to actuate the user’s legs in a motion resembling a standing long jump during teleportation, combined with virtual camera height adjustments

aligned with virtual motion [29, 46]. A standing long jump involves three key movements: knee flexion, knee extension, and ankle flexion. To induce these motions through electrical stimulation, two electrodes were attached to each of the following three leg muscles: the quadriceps, hamstrings, and tibialis anterior (Figure 1 (C)). The quadriceps, located in the anterior thigh, induce knee extension during contraction [17], while the hamstrings, located in the posterior thigh, are associated with knee flexion [35]. Finally, the tibialis anterior, the main ankle dorsiflexor located in the shin [62], contributes to ankle flexion.

Each teleportation is divided into five phases, analogous to the phases of a standing long jump [44, 65]. Haptic feedback through EMS is applied to replicate the corresponding leg movements during each phase, as follows:

Idle-state. This phase refers to the point prior to the initiation of teleportation, during which no jumping motion has been executed (Figure 2 (A)). No feedback is provided at this point.

Preparation. This phase begins when the user presses the controller button to designate the target position, analogous to the point in a standing long jump in which the legs bend for preparation of push-off (Figure 2 (B)). EMS is applied to the hamstrings and tibialis anterior, with intensity linearly adjusted based on the distance to the target, reflecting the joint angle changes in a standing long jump [65]. For example, as the target marker moves farther, the intensity increases, inducing the user's legs to bend further. Simultaneously, the virtual camera height decreases proportionally (up to 0.6m) to simulate the lowered gaze caused by leg flexion. Feedback begins when the target marker is generated and is maintained until the release of the controller button.

Push-off. This phase occurs immediately after the controller button is released to initiate teleportation, corresponding to the point of a standing long jump where the legs extend forcefully to propel the body upward (Figure 2 (C)). EMS is applied to the quadriceps, straightening the user's legs, with intensity determined based on the finalized travel distance. For visual feedback, when the controller button is released, the virtual camera height temporarily rises (up to 0.5m) to represent the vertical motion of leaping into the air, while a fade-out animation is applied to signal teleportation.

Float. This phase corresponds to the floating phase of a standing long jump, during which the body is in mid-air (Figure 2 (D)). During this phase, no stimulation is provided to any muscle group to convey the sensation of floating.

Landing. This phase refers to the point at which teleportation has been completed, corresponding to the landing action in a standing long jump where the ankles flex upon contact with the ground (Figure 2 (E)). EMS is applied to the tibialis anterior using maximum intensity to simulate the ground's impact. Visually, a fade-in animation is applied, with the elevated gaze during the push-off phase returning to the original height.

Considering the time needed for users to perceive visual feedback and the muscle contraction time caused by EMS, a single jump of TeleHopper takes approximately 1 sec from Push-off to Landing (Push-off phase: 0.65 sec, Float phase: 0.1 sec, Landing phase: 0.2 sec).

4 Evaluation

In this study, we aimed to enhance the teleportation experience through proprioceptively simulating the sense of teleportation by providing EMS-based haptic feedback. Accordingly, we conducted a within-subjects study to evaluate users' sense of presence and workload through the application of TeleHopper.

4.1 Experimental setting

4.1.1 Study design. The experimental conditions were established according to the method of haptic stimulation control, resulting in three conditions. Baseline (BL), the first condition, involved teleportation without any haptic feedback. The second condition, labeled as static stimulation (ST), delivered EMS at a constant intensity (the mean value between the minimum and maximum thresholds) regardless of travel distance. This was intended to investigate the effect of varying stimulation intensity based on travel distance, drawing on prior studies that employed similar experimental conditions [9, 23]. The third condition involved the application of our system, TeleHopper (TH). To reduce the impact of visual feedback, consistency in visual cues (including teleportation time) was maintained across conditions. The order of conditions was counterbalanced using a Latin square design.

4.1.2 Task setup. To ensure a diverse range of locomotion experiences, we employed a randomly generated coin collection task. At the start of each trial, a coin was spawned in a random direction at a distance of 20–50 meters from the participant. Participants rotated their bodies to locate the target and teleported to the coin's position. When arriving within a 1.5 meter radius of the coin, participants pressed a button on the controller to collect it, receiving an audio cue as feedback. The virtual environment for the task consisted of a flat terrain surrounded by hills, with obstacles (e.g., ruins) to provide variance in locomotion. A *Quest 3* VR headset was used for the experiment.

4.2 Procedure

The experiment was conducted with 16 participants (11 male, 5 female, $M = 24.9$, $SD = 2.7$). Among them, 15 had prior experience with VR, and 10 had prior experience using EMS. None of the participants reported any contraindications related to EMS (e.g., cardiovascular diseases). The study was approved by the Institutional Review Board.

Before the experiment, participants completed a demographic questionnaire and proceeded with EMS personalization as described in Section 3.1. Participants were given practice time to freely explore the environment to familiarize themselves with the EMS feedback (approximately 5 minutes). Once preparations were complete, participants performed trials under each condition in sequence, collecting coins for 5 minutes per condition. Participants were allowed to request breaks if they experienced fatigue or discomfort at any time during or between trials.

Between conditions, participants completed questionnaires in regards to presence and workload to evaluate their experience for each condition. The questionnaires included the Igroup Presence Questionnaire (IPQ) [52] and the NASA Task Load Index (NASA-TLX) [19]. The IPQ was used to assess the impact of TeleHopper

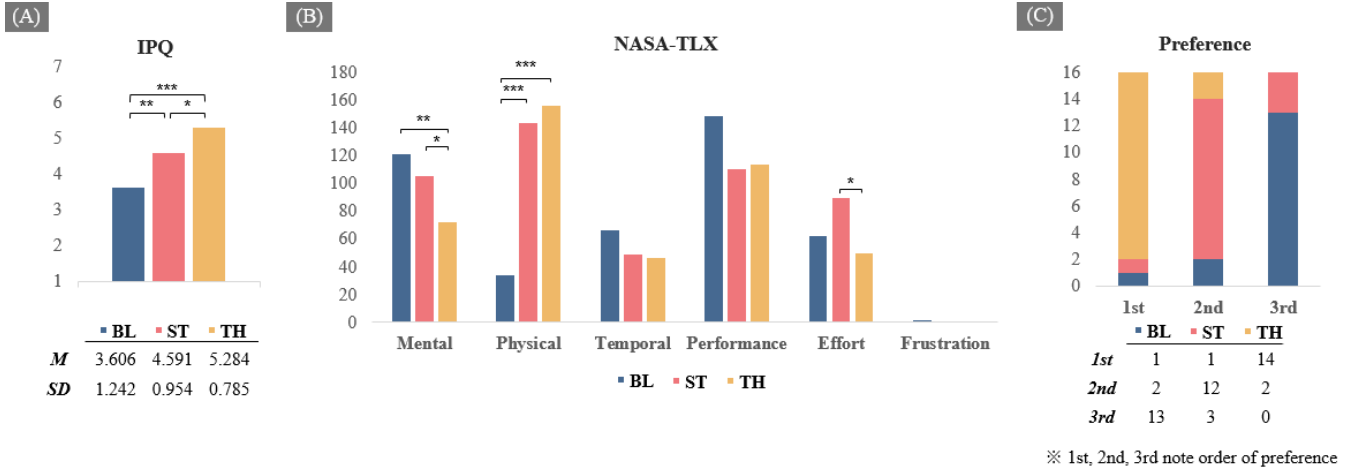


Figure 3: Results of the (A) IPQ questionnaire, (B) NASA-TLX questionnaire, and (C) preference ranking for all conditions. Significance levels are indicated: $p < 0.05^*$, $p < 0.01^{}$, $p < 0.001^{***}$.**

on participants' subjective sense of presence in the virtual environment [42, 53], while the NASA-TLX was used to measure perceived task demands associated with locomotion and virtual environment navigation under the presented stimulation conditions [32, 36]. After the final condition, participants took part in a 10-minute semi-structured interview for qualitative feedback in regards to their experience in each condition, and were asked to rank the three conditions in order of overall preference.

5 Results

The results for the IPQ and NASA-TLX questionnaires were tested for normality based on skewness (< 3.0) and kurtosis (< 10.0) [28]. A repeated measures ANOVA was then conducted, and if statistical significance ($p < 0.05^*$) was found, Bonferroni correction was applied for post-hoc analysis.

5.1 Presence

Presence scores were calculated by averaging all items in the IPQ. Significant differences were observed across the three conditions ($F(1.381, 20.721) = 23.615$, $p < 0.001^{***}$) (Figure 3 (A)). TeleHopper significantly increased presence compared to both the static stimulation and the baseline condition ($p = 0.025^*$, $p < 0.001^{***}$, respectively). Additionally, the static stimulation condition was rated significantly higher than the baseline condition ($p = 0.001^{**}$).

5.2 Workload

To analyze perceived workload, we combined the raw and paired-choice data for each of the six subscales of the NASA-TLX and calculated the weighted results. While the total NASA-TLX score did not show significant differences (see Appendix A), the survey subscales revealed significant differences in mental demand ($F(2, 30) = 7.491$, $p = 0.002^{**}$), physical demand ($F(2, 30) = 18.317$, $p < 0.001^{***}$), and effort ($F(1.408, 21.114) = 4.507$, $p = 0.034^*$) (Figure 3 (B)). For mental demand, TeleHopper was rated significantly lower than both the static stimulation condition ($p = 0.046^*$) and the baseline condition ($p = 0.002^{**}$). Conversely, in terms of physical

demand, both TeleHopper and the static stimulation condition were significantly higher than the baseline condition (both $p < 0.001^{***}$). Regarding effort, TeleHopper was assessed to be significantly lower than the static stimulation condition ($p = 0.019^*$). No significant differences were observed for the remaining subscales.

5.3 Qualitative Feedback

Regarding preference, most participants selected TeleHopper as their most preferred condition, followed by the static stimulation condition and baseline condition respectively (Figure 3 (C)). Participants found TeleHopper's variation in stimulus intensity realistic and immersive ($P1-3$, 7, 11-14), noting that the alignment of imagined and real bodily sensations provided spatial movement ($P1$, 4-5, 8-9, 15-16). Some also reported a sense of agency, describing it as "feeling like I am taking action" ($P1$, 9). Participants who did not rank TeleHopper highest mentioned the influence of uncomfortable tingling associated with EMS ($P6$, 10). The reasoning for static stimulation being less preferred than TeleHopper was that its constant nature was perceived as "less realistic" and "boring" ($P1-5$, 7-15). Finally, regarding the baseline, participants described it as easily becoming monotonous and lacking realism ($P1-5$, 7-16). Only one participant ranked the baseline condition as most preferred, explaining that "the absence of stimulation felt comfortable" ($P6$).

We also asked participants about the alignment of TeleHopper's feedback with teleportation. Most felt it was "well-suited to the virtual locomotion mechanism of teleportation" ($P1-3$, 5-8, 10, 12-13, 15-16). However, some felt the feedback did not align well with teleportation, describing it as "feeling more like jumping in place" ($P9$, 11, 14). Others disassociated the sensation of jumping from teleportation, with opinions that teleportation should feel more like "evoking a flying sensation" ($P4$) or "the body being distorted" ($P9$). Additional comments regarding TeleHopper's feedback design include that the variations in stimulation intensity helped them estimate travel distances ($P1$, 4-5, 8-9, 11-13, 16). Specifically, some participants stated that "the other two conditions lacked

distance-related cues, resulting in relatively higher mental load during locomotion” (P1, 16).

6 Discussion

The study results indicate that TeleHopper provided a generally positive user experience, demonstrated by increased presence and reduced mental demand. The significant improvement in presence observed with TeleHopper compared to the baseline condition is consistent with prior research, which highlights the positive influence of proprioception on presence [14]. Through TeleHopper’s EMS-induced muscle contractions facilitating joint rotation at the knees and ankles, the activation of movement of the surrounding tendons, skin, and muscles effectively delivered the proprioceptive sensation of jumping, thereby enhancing the sense of presence. Additionally, for TeleHopper, the kinetic movements generated were proportional to travel distance, leading to a significant enhancement of presence compared to the static stimulation. This suggests that the proprioceptive feedback provided by EMS not only acted as a teleportation event cue, but also conveyed varying levels of muscle contraction and tensile strength. These variations allowed users to intuitively discern the magnitude of movement required, reinforcing their sense of agency and their ability to perceive the journey to the target location. These findings are consistent with previous studies indicating that the human neuromuscular system efficiently regulates muscle recruitment in response to movement difficulty [5, 54]. In contrast, the constant EMS intensity in the static stimulation condition, regardless of visually perceived travel distance, potentially created a mismatch between visual perception and proprioception, resulting in reduced presence. This interpretation aligns with feedback from 87.5% of participants, who described the static stimulation condition as “less realistic” (P1-5, 7-15).

Both conditions involving proprioceptive feedback were rated as having higher physical demand compared to the baseline condition. This outcome is expected, given that electrical stimulation directly affects muscle activation and subsequent kinesthetic motion [8]. However, TeleHopper demonstrated notably lower mental demand than the other conditions, suggesting that proprioceptive feedback aligned with travel distance plays a key role in reducing cognitive load. Navigating virtual environments inherently imposes cognitive demands, requiring users to constantly monitor their position and determine their direction of movement [7, 36]. In this context, TeleHopper’s support for distance perception likely alleviated some of these demands by providing proprioceptive feedback. This interpretation aligns with participant responses, as more than half (56.25%; P1, 4-5, 8-9, 11-13, 16) reported that variations in EMS intensity helped them estimate travel distances. Additional sensory input on TeleHopper reduced reliance on visual cues alone, potentially explaining the observed decrease in overall mental effort compared to conditions without such feedback. An additional notable finding is that despite the higher physical demand, 87.5% of participants (14 out of 16) identified TeleHopper as their most preferred condition. This indicates that users valued the benefits of enhanced presence and reduced mental demand, viewing the additional physical effort from EMS as an acceptable trade-off.

Limitations and future work. A variety of future research avenues can be pursued to further improve and expand upon the

potential of simulating proprioceptive feedback for teleportation. For instance, TeleHopper currently provides stimulation variations only for horizontal locomotion. Since jumping to different heights in real life distinctly alters the proprioceptive sensations experienced, applying realistic proprioceptive feedback to include vertical as well as horizontal movement could enhance teleportation in scenarios with elevated terrain or aerial environments [41, 60]. In addition, to directly induce user movement and enhance proprioception, this study applied EMS exclusively to the legs. Future exploration could involve the targeting of other body parts or combining with alternative haptic feedback methods. For instance, the addition of vibrotactile feedback applied to the soles of the feet could enrich the teleportation experience by simulating sensations such as foot pressure or landing impact [29, 40]. Furthermore, to ensure participant safety, this study applied only a stimulation intensity sufficient to induce a motion similar to a jump, rather than having users perform an actual leap. For a more biomechanically accurate experience, additional haptic feedback associated with jumping may be applied, such as simulating tension in the ankle tendons [45] or providing force feedback to simulate body weight [43].

Additionally, some participants showed differing opinions on what teleportation should physically feel like, as opposed to jumping. Considering that mental representations of actions or concepts are grounded in physical experiences, such diverse interpretations of teleportation sensation among participants suggest that feedback perception is shaped by individual cognitive processing styles [3, 13]. This warrants further investigation into integrating a wider variety of physical sensations correlating with teleportation, such as simulating flight or distortion. Finally, EMS’s reliance on muscle activation and subsequent involuntary motion may affect user experience during prolonged use. Electrical stimulation through the skin can cause discomfort [56], such as the tingling sensation reported during participant interviews, while continuous muscle activation may lead to fatigue. These factors likely contributed to the higher physical load of TeleHopper and could become further intensified during prolonged use, potentially impacting performance and usability. Therefore, future research should explore the long-term effects of EMS on TeleHopper through prolonged trials or comparative analyses based on participants’ prior exposure to EMS.

7 Conclusion

We propose TeleHopper, a system that supplements the lack of proprioceptive sensation in teleportation by simulating the sense of jumping through EMS. The feedback design replicates the phases of a standing long jump by targeting key muscle groups to induce motions such as knee flexion, knee extension, and ankle flexion. In a user study comparing TeleHopper with baseline and static stimulation conditions, TeleHopper significantly enhanced presence and reduced mental load compared to the other conditions, by providing realistic proprioception and distance estimation through stimulation intensity variations based on travel distance. Future work could explore extending stimulation variations for vertical movement, integrating other body parts or combining with alternative haptic feedback methods.

Acknowledgments

This work was partly supported by the Institute of Information & Communications Technology Planning & Evaluation (IITP)-ITRC (Information Technology Research Center) grant funded by the Korea government (MSIT)(IITP-2025-RS-2024-00436398, 33%), the Gwangju Institute of Science and Technology (GIST)–Massachusetts Institute of Technology (MIT) Research Collaboration Grant in 2025 (33%), and the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (RS-2024-00343397) (33%).

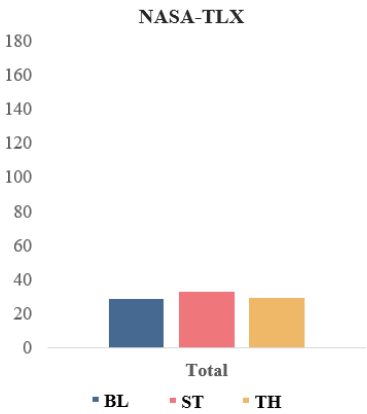
References

- [1] Ashu Adhikari, Daniel Zielasko, Ivan Aguilar, Alexander Bretin, Ernst Kruijff, Markus von der Heyde, and Bernhard E Riecke. 2022. Integrating continuous and teleporting vr locomotion into a seamless ‘hyperjump’ paradigm. *IEEE Transactions on Visualization and Computer Graphics* 29, 12 (2022), 5265–5281. doi:10.1109/TVCG.2022.3207157
- [2] Niels H Bakker, Peter O Passenier, and Peter J Werkhoven. 2003. Effects of head-slaved navigation and the use of teleports on spatial orientation in virtual environments. *Human factors* 45, 1 (2003), 160–169. doi:10.1518/hfes.45.1.160.27234
- [3] Lawrence W Barsalou, W Kyle Simmons, Aron K Barbey, and Christine D Wilson. 2003. Grounding conceptual knowledge in modality-specific systems. *Trends in cognitive sciences* 7, 2 (2003), 84–91. doi:10.1016/S1364-6613(02)00029-3
- [4] Jiwan Bhandari, Paul R MacNeilage, and Eelke Folmer. 2018. Teleportation without Spatial Disorientation Using Optical Flow Cues. In *Graphics interface*. Canadian Human-Computer Communications Society, 162–167. doi:10.20380/gi2018.22
- [5] Caroline Blanchard, Régine Roll, Jean-Pierre Roll, and Anne Kavounoudias. 2013. Differential contributions of vision, touch and muscle proprioception to the coding of hand movements. *PLoS one* 8, 4 (2013), e62475. doi:10.1371/journal.pone.0062475
- [6] Costas Boletsis and Jarl Erik Cedergrén. 2019. VR locomotion in the new era of virtual reality: an empirical comparison of prevalent techniques. *Advances in Human-Computer Interaction* 2019, 1 (2019), 7420781. doi:10.1155/2019/7420781
- [7] Doug A Bowman, David Koller, and Larry F Hodges. 1997. Travel in immersive virtual environments: An evaluation of viewpoint motion control techniques. In *Proceedings of IEEE 1997 Annual International Symposium on Virtual Reality*. IEEE, 45–52. doi:10.1109/VRRAIS.1997.583043
- [8] Fabio Buttussi and Luca Chittaro. 2019. Locomotion in place in virtual reality: A comparative evaluation of joystick, teleport, and leaning. *IEEE transactions on visualization and computer graphics* 27, 1 (2019), 125–136. doi:10.1109/TVCG.2019.2928304
- [9] Shao-Yu Chu, Yun-Ting Cheng, Shih Chin Lin, Yung-Wen Huang, Yi Chen, and Mike Y Chen. 2021. Motionring: Creating illusory tactile motion around the head using 360 vibrotactile headbands. In *The 34th Annual ACM Symposium on User Interface Software and Technology*. ACM, 724–731. doi:10.1145/3472749.3474781
- [10] Jeremy Clifton and Stephen Palmisano. 2020. Effects of steering locomotion and teleporting on cybersickness and presence in HMD-based virtual reality. *Virtual Reality* 24, 3 (2020), 453–468. doi:10.1007/s10055-019-00407-8
- [11] Sebastian Cmentowski, Andrey Krekhov, and Jens Krüger. 2019. Outstanding: A multi-perspective travel approach for virtual reality games. In *Proceedings of the annual symposium on computer-human interaction in play*. ACM, 287–299. doi:10.1145/3311350.3347183
- [12] Ahmed Ibrahim Ahmed Mohamed Elsharkawy, Aya Abdunnasser Saed Ataya, Dohyeon Yeo, Eunsol An, Seokhyun Hwang, and SeungJun Kim. 2024. SYNC-VR: Synchronizing Your Senses to Conquer Motion Sickness for Enriching In-Vehicle Virtual Reality. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM, 1–17. doi:10.1145/3613904.3642941
- [13] Lucia Foglia and Robert A Wilson. 2013. Embodied cognition. *Wiley Interdisciplinary Reviews: Cognitive Science* 4, 3 (2013), 319–325. doi:10.1002/wcs.1226
- [14] Jann Philipp Freiwald, Oscar Ariza, Omar Janeh, and Frank Steinicke. 2020. Walking by cycling: A novel in-place locomotion user interface for seated virtual reality experiences. In *Proceedings of the 2020 CHI conference on human factors in computing systems*. ACM, 1–12. doi:10.1145/3313831.3376574
- [15] Bocheon Gim, Seokhyun Hwang, Seongjun Kang, Gwangbin Kim, Dohyeon Yeo, and Seungjun Kim. 2025. I Want to Break Free: Enabling User-Applied Active Locomotion in In-Car VR through Contextual Cues. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM, 1–19. doi:10.1145/3706598.3713373
- [16] Bocheon Gim, Seongjun Kang, Gwangbin Kim, Dohyeon Yeo, Seokhyun Hwang, and Seungjun Kim. 2024. Curving the Virtual Route: Applying Redirected Steering Gains for Active Locomotion in In-Car VR. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. ACM, 1–7. doi:10.1145/3613905.3650746
- [17] Edward S Grood, Wilfredo J Suntay, FRANK R Noyes, and David L Butler. 1984. Biomechanics of the knee-extension exercise. Effect of cutting the anterior cruciate ligament. *JBJS* 66, 5 (1984), 725–734. doi:10.2106/00004623-198466050-00011
- [18] Ping-Hsuan Han, Yang-Sheng Chen, Kong-Chang Lee, Hao-Cheng Wang, Chiao-En Hsieh, Jui-Chun Hsiao, Chien-Hsing Chou, and Yi-Ping Hung. 2018. Haptic around: multiple tactile sensations for immersive environment and interaction in virtual reality. In *Proceedings of the 24th ACM symposium on virtual reality software and technology*. ACM, 1–10. doi:10.1145/3281505.3281507
- [19] Sandra G. Hart. 1988. Development of NASA-TLX (Task Load Index): Results of Empirical and Theoretical Research. In *Human Mental Workload*. Elsevier, 139–183. doi:10.1016/S0166-4115(08)62386-9
- [20] Emil R Hoeg, Kevin V Ruder, Niels C Nilsson, Rolf Nordahl, and Stefania Serafin. 2017. An exploration of input conditions for virtual teleportation. In *2017 IEEE Virtual Reality (VR)*. IEEE, 341–342. doi:10.1109/VR.2017.7892316
- [21] Seokhyun Hwang, Seongjun Kang, Jeongseok Oh, Jeongju Park, Semoo Shin, Yiyue Luo, Joseph DelPreto, Sangbeom Lee, Kyoobin Lee, Wojciech Matusik, Daniela Rus, and SeungJun Kim. 2025. TelePulse: Enhancing the Teleoperation Experience through Biomechanical Simulation-Based Electrical Muscle Stimulation in Virtual Reality. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM, 1–26. doi:10.1145/3706598.3713767
- [22] Seokhyun Hwang, Jieun Lee, Youngin Kim, Youngseok Seo, and Seungjun Kim. 2023. Electrical, vibrational, and cooling stimuli-based redirected walking: Comparison of various vestibular stimulation-based redirected walking systems. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. ACM, 1–18. doi:10.1109/ISMAR59233.2023.00135
- [23] Seokhyun Hwang, Jeongseok Oh, Seongjun Kang, Minwoo Seong, Ahmed Ibrahim Ahmed Mohamed Elsharkawy, and SeungJun Kim. 2024. Ergopulse: Electrifying your lower body with biomechanical simulation-based electrical muscle stimulation haptic system in virtual reality. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM, 1–21. doi:10.1145/3613904.3642008
- [24] Seongjun Kang, Gwangbin Kim, Seokhyun Hwang, Jeongju Park, Ahmed Ibrahim Ahmed Mohamed Elsharkawy, and SeungJun Kim. 2024. Flip-Pelt: Motor-Driven Peltier Elements for Rapid Thermal Stimulation and Congruent Pressure Feedback in Virtual Reality. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*. ACM, 1–15. doi:10.1145/3654777.3676363
- [25] Seongjun Kang, Gwangbin Kim, and SeungJun Kim. 2023. Giant Finger: A Novel Visuo-Somatosensory Approach to Simulating Lower Body Movements in Virtual Reality. In *2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE, 233–242. doi:10.1109/ISMAR59233.2023.00038
- [26] Mohamed Khamis, Nora Schuster, Ceenu George, and Max Pfeiffer. 2019. Electrocusenes: Realistic haptic feedback in cutscenes of virtual reality games using electric muscle stimulation. In *Proceedings of the 25th ACM Symposium on Virtual Reality Software and Technology*. ACM, 1–10. doi:10.1145/3359996.3364250
- [27] Jinwook Kim, Seonghyeon Kim, and Jeongmi Lee. 2022. The effect of multisensory pseudo-haptic feedback on perception of virtual weight. *IEEE Access* 10 (2022), 5129–5140. doi:10.1109/ACCESS.2022.3140438
- [28] Rex B Kline. 2023. *Principles and practice of structural equation modeling*. Guilford publications.
- [29] Ernst Kruijff, Alexander Marquardt, Christina Treppkowski, Robert W Lindeman, Andre Hinkenjann, Jens Maiero, and Bernhard E Riecke. 2016. On your feet! enhancing vection in leaning-based interfaces through multisensory stimuli. In *Proceedings of the 2016 Symposium on Spatial User Interaction*. ACM, 149–158. doi:10.1145/2983310.2985759
- [30] Lucie Kruse, Sungchul Jung, Richard Chen Li, and Robert W Lindeman. 2020. On the Use of Jumping Gestures for Immersive Teleportation in VR. In *ICAT-EGVE*. The Eurographics Association, 113–120. doi:10.2312/egve.20201265
- [31] Eike Langbehn, Paul Lubos, and Frank Steinicke. 2018. Evaluation of locomotion techniques for room-scale vr: Joystick, teleportation, and redirected walking. In *Proceedings of the Virtual Reality International Conference-Laval Virtual*. IEEE, 1–9. doi:10.1145/3234253.3234291
- [32] Jieun Lee, Seokhyun Hwang, Aya Ataya, and SeungJun Kim. 2024. Effect of optical flow and user VR familiarity on curvature gain thresholds for redirected walking. *Virtual Reality* 28, 1 (2024), 35. doi:10.1007/s10055-023-00935-4
- [33] Jieun Lee, Seokhyun Hwang, Kyunghwan Kim, and SeungJun Kim. 2024. Evaluation of visual, auditory, and olfactory stimulus-based attractors for intermittent reorientation in virtual reality locomotion. *Virtual Reality* 28, 2 (2024), 104. doi:10.1007/s10055-024-00997-y
- [34] Daniele Leonardi, Antonio Frisoli, Michele Barsotti, Marcello Carrozzino, and Massimo Bergamasco. 2014. Multisensory feedback can enhance embodiment within an enriched virtual walking scenario. *Presence* 23, 3 (2014), 253–266. doi:10.1162/PRES_a_00190
- [35] G Li, TW Rudy, M Sakane, A Kanamori, CB Ma, and SL-Y Woo. 1999. The importance of quadriceps and hamstring muscle loading on knee kinematics and in-situ forces in the ACL. *Journal of biomechanics* 32, 4 (1999), 395–400. doi:10.1016/S0021-9290(98)00181-X
- [36] Donghae Lim, Shizuka Shirai, Jason Orlosky, Photchara Ratsamee, Yuki Uranishi, and Haruo Takemura. 2022. Evaluation of user interfaces for three-dimensional

- locomotion in virtual reality. In *Proceedings of the 2022 ACM Symposium on Spatial User Interaction*. ACM, 1–9. doi:10.1145/3565970.3567693
- [37] Pedro Lopes and Patrick Baudisch. 2017. Immense power in a tiny package: Wearables based on electrical muscle stimulation. *IEEE Pervasive Computing* 16, 3 (2017), 12–16. doi:10.1109/MPRV.2017.2940953
- [38] Pedro Lopes, Alexandra Ion, and Patrick Baudisch. 2015. Impacto: Simulating physical impact by combining tactile stimulation with electrical muscle stimulation. In *Proceedings of the 28th annual ACM symposium on user interface software & technology*. ACM, 11–19. doi:10.1145/2807442.2807443
- [39] Pedro Lopes, Sijing You, Lung-Pan Cheng, Sebastian Marwecki, and Patrick Baudisch. 2017. Providing haptics to walls & heavy objects in virtual reality by means of electrical muscle stimulation. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. ACM, 1471–1482. doi:10.1145/3025453.3025600
- [40] Yusuke Matsuda, Junya Nakamura, Tomohiro Amemiya, Yasushi Ikei, and Michiteru Kitazaki. 2021. Enhancing virtual walking sensation using self-avatar in first-person perspective and foot vibrations. *Frontiers in Virtual Reality* 2 (2021), 654088. doi:10.3389/frvir.2021.654088
- [41] Andrii Matvienko, Florian Müller, Martin Schmitz, Marco Fendrich, and Max Mühlhäuser. 2022. Skyport: Investigating 3d teleportation methods in virtual environments. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. ACM, 1–11. doi:10.1145/3491102.3501983
- [42] Florian Müller, Arantxa Ye, Dominik Schön, and Julian Rasch. 2023. UndoPort: Exploring the Influence of Undo-Actions for Locomotion in Virtual Reality on the Efficiency, Spatial Understanding and User Experience. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. ACM, 1–15. doi:10.1145/3544548.3581557
- [43] Romain Nith, Jacob Serfaty, Samuel G Shatzkin, Alan Shen, and Pedro Lopes. 2023. JumpMod: Haptic backpack that modifies users' perceived jump. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. ACM, 1–15. doi:10.1145/3544548.3580764
- [44] Yu Okubo, Koji Kaneoka, Itsuo Shiina, Masaki Tatsumura, and Shumpei Miyakawa. 2013. Abdominal muscle activity during a standing long jump. *Journal of Orthopaedic & Sports Physical Therapy* 43, 8 (2013), 577–582. doi:10.2519/jospt.2013.4420
- [45] Takashi Ota, Keigo Matsumoto, Kazuma Aoyama, Tomohiro Amemiya, Takuji Narumi, and Hideaki Kuzuoka. 2024. Ankle tendon electrical stimulation to enhance sensation of walking on a slope in walking-in-place. *Frontiers in Virtual Reality* 5 (2024), 1456202. doi:10.3389/frvir.2024.1456202
- [46] Stephen Palmisano, Juno Kim, Robert Allison, and Frederick Bonato. 2011. Simulated viewpoint jitter shakes sensory conflict accounts of vection. *Seeing and perceiving* 24, 2 (2011), 173–200. doi:10.1163/187847511X57087
- [47] Max Pfeiffer, Tim Diente, and Michael Rohs. 2016. Let your body move: a prototyping toolkit for wearable force feedback with electrical muscle stimulation. In *Proceedings of the 18th International Conference on Human-Computer Interaction with Mobile Devices and Services*. ACM, 418–427. doi:10.1145/2935334.2935348
- [48] Uwe Proske and Simon C Gandevia. 2012. The proprioceptive senses: their roles in signaling body shape, body position and movement, and muscle force. *Physiological Reviews* 92, 4 (2012), 1651–1697. doi:10.1152/physrev.00048.2011
- [49] Serge Rossignol, Réjean Dubuc, and Jean-Pierre Gossard. 2006. Dynamic sensorimotor interactions in locomotion. *Physiological reviews* 86, 1 (2006), 89–154. doi:10.1152/physrev.00028.2005
- [50] Ehsan Sayyad, Misha Sra, and Tobias Höllerer. 2020. Walking and teleportation in wide-area virtual reality experiences. In *2020 IEEE international symposium on mixed and augmented reality (ISMAR)*. IEEE, 608–617. doi:10.1109/ISMAR50242.2020.00088
- [51] Alexander Schäfer, Gerd Reis, and Didier Stricker. 2021. Controlling teleportation-based locomotion in virtual reality with hand gestures: a comparative evaluation of two-handed and one-handed techniques. *Electronics* 10, 6 (2021), 715. doi:10.3390/electronics10060715
- [52] Thomas Schubert, Frank Friedmann, and Holger Regenbrecht. 2001. The experience of presence: Factor analytic insights. *Presence: Teleoperators & Virtual Environments* 10, 3 (2001), 266–281. doi:10.1162/105474601300343603
- [53] Mel Slater, Anthony Steed, John McCarthy, and Francesco Maringelli. 1998. The influence of body movement on subjective presence in virtual environments. *Human factors* 40, 3 (1998), 469–477. doi:10.1518/001872098779591368
- [54] Samuel J Sober and Philip N Sabes. 2003. Multisensory integration during motor planning. *Journal of Neuroscience* 23, 18 (2003), 6982–6992. doi:10.1523/JNEUROSCI.23-18-06982.2003
- [55] Primož Strojnik, Alojz Kralj, and I Ursic. 1979. Programmed six-channel electrical stimulator for complex stimulation of leg muscles during walking. *IEEE Transactions on Biomedical Engineering* BME-26, 2 (1979), 112–116. doi:10.1109/TBME.1979.326520
- [56] Yudai Tanaka, Akifumi Takahashi, and Pedro Lopes. 2023. Interactive Benefits from Switching Electrical to Magnetic Muscle Stimulation. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. ACM, 1–12. doi:10.1145/3586183.3606812
- [57] Chun-Miao Tseng, Po-Yu Chen, Shih Chin Lin, Yu-Wei Wang, Yu-Hsin Lin, Mu-An Kuo, Neng-Hao Yu, and Mike Y Chen. 2022. Headwind: Enhancing teleportation experience in vr by simulating air drag during rapid motion. In *Proceedings of the 2022 CHI conference on human factors in computing systems*. ACM, 1–11. doi:10.1145/3491102.3501890
- [58] Juwon Um, Eunki Jeon, Yumin Kang, Seongjun Kang, Ahmed Elsharkawy, Joseph DelPreto, Wojciech Matusik, Daniela Rus, and SeungJun Kim. 2024. LegSense: Inducing Walking Sensation in Seated VR by Providing Movement Illusion via Electrical Muscle Stimulation. In *Companion of the 2024 on ACM International Joint Conference on Pervasive and Ubiquitous Computing*. ACM, 797–802. doi:10.1145/3675094.3678375
- [59] Julius Von Willich, Martin Schmitz, Florian Müller, Daniel Schmitt, and Max Mühlhäuser. 2020. Podoportation: Foot-based locomotion in virtual reality. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. ACM, 1–14. doi:10.1145/3313831.3376626
- [60] Tim Weissker, Pauline Bimberg, Aalok Shashidhar Gokhale, Torsten Kühlen, and Bernd Froehlich. 2023. Gaining the high ground: Teleportation to mid-air targets in immersive virtual environments. *IEEE Transactions on Visualization and Computer Graphics* 29, 5 (2023), 2467–2477. doi:10.1109/TVCG.2023.3247114
- [61] Tim Weißker, André Kunert, Bernd Fröhlich, and Alexander Kulik. 2018. Spatial updating and simulator sickness during steering and jumping in immersive virtual environments. In *2018 IEEE conference on virtual reality and 3D user interfaces (VR)*. IEEE, 97–104. doi:10.1109/VR.2018.8446620
- [62] Michael W Whittle. 2014. *Gait analysis: an introduction*. Butterworth-Heinemann.
- [63] Betsy Williams, Gayathri Narasimham, Bjoern Rump, Timothy P McNamara, Thomas H Carr, John Rieser, and Bobby Bodenheimer. 2007. Exploring large virtual environments with an HMD when physical space is limited. In *Proceedings of the 4th symposium on Applied perception in graphics and visualization*. ACM, 41–48. doi:10.1145/1272582.1272590
- [64] Dennis Wolf, Katja Rogers, Christoph Kunder, and Enrico Rukzio. 2020. Jumpvr: Jump-based locomotion augmentation for virtual reality. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. ACM, 1–12. doi:10.1145/3313831.3376243
- [65] Wen-Lan Wu, Jia-Hroung Wu, Hwai-Ting Lin, and Gwo-Jaw Wang. 2003. Biomechanical analysis of the standing long jump. *Biomedical Engineering: Applications, Basis and Communications* 15, 05 (2003), 186–192. doi:10.4015/S1016237203000286

A Additional NASA-TLX results

(A)



(B)

NASA-TLX	Condition	M	SD
Mental demand	Baseline	121.250	99.121
	Static stimulation	105.625	110.210
	TeleHopper	72.500	90.885
Physical demand	Baseline	33.750	69.174
	Static stimulation	143.750	97.699
	TeleHopper	156.250	102.559
Temporal demand	Baseline	66.250	81.148
	Static stimulation	48.750	76.059
	TeleHopper	46.250	68.105
Performance	Baseline	148.750	141.368
	Static stimulation	110.625	90.662
	TeleHopper	113.750	120.160
Effort	Baseline	62.500	92.268
	Static stimulation	89.375	88.353
	TeleHopper	50.000	51.381
Frustration	Baseline	1.250	5.000
	Static stimulation	0.000	0.000
	TeleHopper	0.000	0.000
Total	Baseline	28.918	20.510
	Static stimulation	33.208	17.950
	TeleHopper	29.250	21.497

Figure 4: (A) Results of the total NASA-TLX score, (B) Descriptive statistics for each of the NASA-TLX subscales.